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328872 (28)

BE (8th Semester)

Examination, April-May, 2014

Branch : Et&T

RADAR ENGINEERING & NAVGATIONAL AIDS (PE-III)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : All units contains equal marks. Part (a) is compulsory. Solve any two parts from (b, c, d).

UNIT - I

Q. 1. (a) What do you mean by integration of Radar Pulses. 2

(b) Derive the expression for Radar equation and what are the limitations of this equation. 7

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(c) A ground-based air-surveillance radar operates at a frequency of 1300 MHz. Its maximum range is 200 nmi for the detection of a target with a radar cross section of one square meter. Its antenna is 12m wide by 4m high, and the antenna aperture efficiency is $\rho_a = 0.65$. The receiver minimum detectable signal is $S_{min} = 10^{-13}$ W

Determine the following :

(i) Antenna effective aperture Ae, and gain of antenna in dB.

(ii) Peak transmitter power

(iii) Pulse repetition freuency to achieve a maximum umambiguous range of 200 nmi.

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(iv) Average transmitter power, if the pulse width is 2 μs

(v) Duty cycle 7

(d) (i) What is the basic principle of Radar (pulsed).

(ii) Applications of Radar. Explain in brief.

(iii) A radar has bandwidth B = 50 KHz and an average time between false alarm of 10 min. Calculate probability of false alarm. 7

UNIT - II

Q. 2. (a) Explain delay line cancellers. 2

(b) (i) A pulse doppler radar has a carrier frequency of 9 GHz and PRF of 4000 Hz.

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Find its blind doppler frequencies and the radial velocity of target which would be undetected by the radar.

What modifications can be made to the radar system (used in part i) that would eliminate blind speed below 2000 m/s and find out the maximum unambigious range of that modified radar system. 7

(c) Compare monopulse amplitude tracking, conical scanning and sequential lobing in briefly. 7

(d) (i) Explain low angle tracking.

(ii) Explain tracking in range. 7

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UNIT - III

Q. 3. (a) What problem occur when propagation of radar waves scattered from Round Earth's Surface. 2

(b) Explain in detail propagation of radar waves in forward scattering from a flat earth. 7

(c) Explain in detail standard propagation and Non-standard propagation. 7

(d) Explain properly the different types of Noise occur which limits the detectability of targets. As tell as other propagation effects. 7

UNIT - IV

Q. 4. (a) What are the parameters of Antenna and explain briefly. 2

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(b) Explain in detail different types of feeds used in parabolic antenna as well as feed support. Also, explain a type of parabolic antenna whose pattern is asymmetric in nature. 7

(c) Write a short note on : 7

(i) Radome

(ii) Cosecant squared pattern

(d) Explain the radiation pattern obtain in phased array antenna. Also, explain different types of shifters used in phased array antenna. 7

UNIT - V

Q. 5. (a) Why is diode limiter following the duplexer sometimes used in receiver protector. 2

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(b) Write a short note on : 7

(i) ECM

(ii) ECCM

(c) Explain in detail different types of display used in Radar. 7

(d) Explain linear beam power tubes and solid state power sources. 7

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